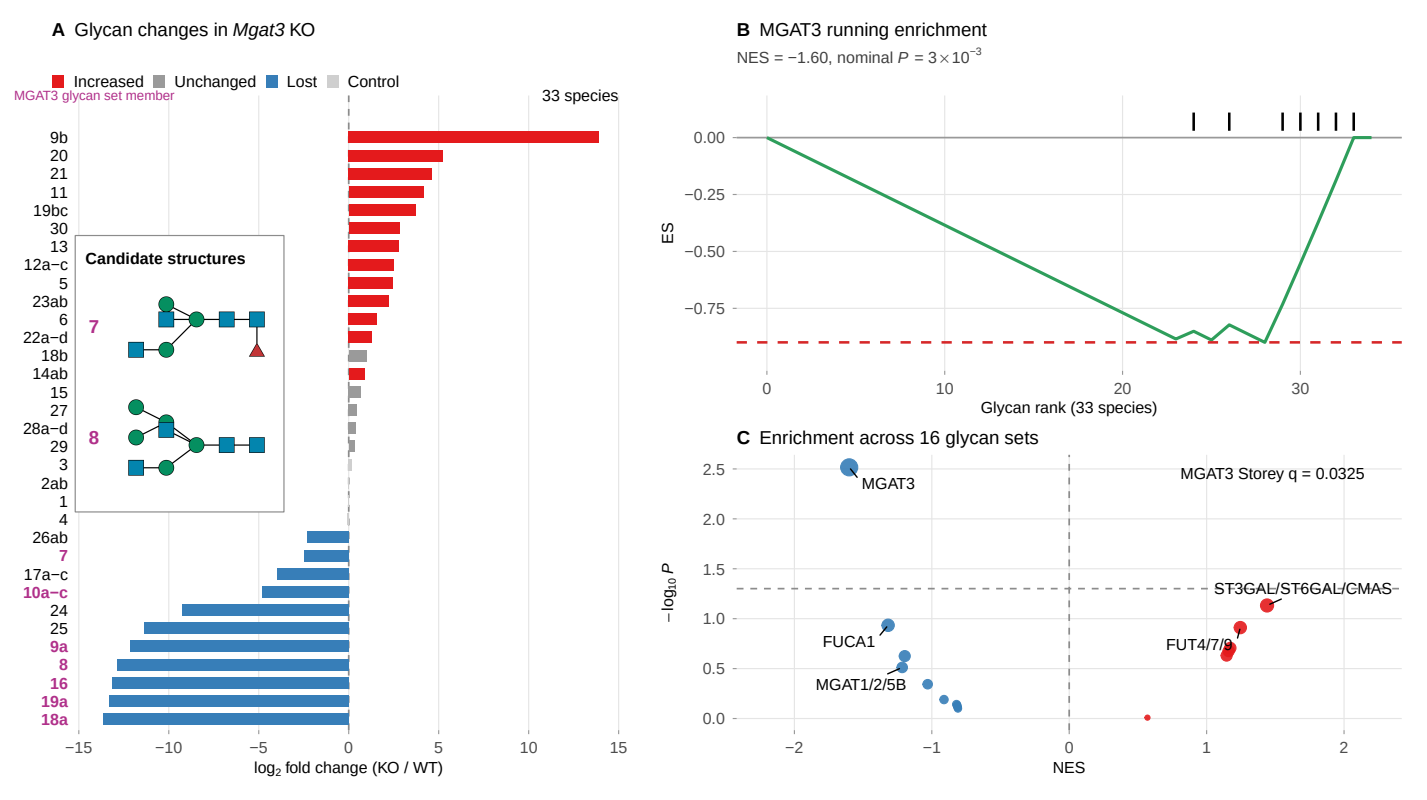

A Glycan changes in Mgat3 KO
Increased
Unchanged
Lost
Control
MGAT3 glycan set member
33 species
9b
20
21
11
19bc
30
13
12a−c
5
23ab
6
22a−d
18b
14ab
15
27
28a−d
29
3
2ab
1
4
26ab
7
17a−c
10a−c
24
25
9a
8
16
19a
18a
Candidate structures
7
8
−15
−10
−5
0
5
10
15
log2 fold change (KO / WT)
B MGAT3 running enrichment
NES = −1.60, nominal P = 3×10−3
0.00
−0.25
−0.50
−0.75
ES
0
10
20
30
Glycan rank (33 species)
C Enrichment across 16 glycan sets
MGAT3 Storey q = 0.0325
MGAT3
ST3GAL/ST6GAL/CMAS
FUCA1
FUT4/7/9
MGAT1/2/5B
2.5
2.0
1.5
1.0
0.5
0.0
−log10 P
−2
−1
0
1
2
NES